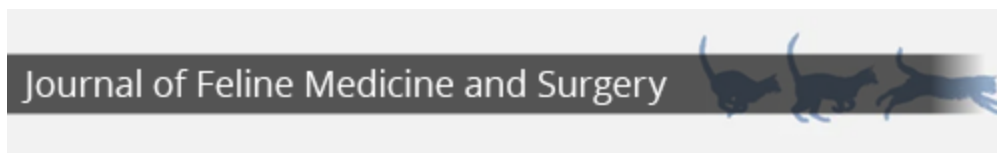


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Serum vitamin D metabolites do not differ between cats with chronic kidney disease and healthy cats

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Abstract

Objective

The aim of the present study was to characterize comprehensive vitamin D metabolite profiles in cats with chronic kidney disease (CKD) compared with healthy cats.

Methods

A total of 33 healthy cats and 66 cats with CKD (six International Renal Interest Society [IRIS] stage 1, 36 IRIS stage 2, 18 IRIS stage 3, six IRIS stage 4) were prospectively enrolled. Serum vitamin D metabolites, including 25-hydroxyvitamin D₃ (25[OH]D₃), 3-epi-25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), 1,25-dihydroxyvitamin D₃ (1,25[OH]₂D₃; calcitriol) and 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃), were analyzed via liquid chromatography-tandem mass spectrophotometry. Vitamin D metabolites, clinicopathologic parameters, age, body weight and

body condition score were compared between groups using a Mann–Whitney U-test or *t*-test as appropriate. Correlations between vitamin D metabolites and clinicopathologic parameters, age and weight were assessed using Spearman's rank test.

Results

There was no significant difference in any vitamin D metabolite between groups. Median 25(OH)D₃ concentrations in cats with CKD and healthy cats were 28.9 ng/ml (range 12.2–68.8) and 26.3 ng/ml (range 11.5–55.9), respectively. Median 3-epi-25(OH)D₃ concentrations in cats with CKD and healthy cats were 9.1 ng/ml (range 2.2–30.4) and 7.2 ng/ml (range 1.9–22.9), respectively. Median calcitriol concentrations in cats with CKD and healthy cats were 141.3 pg/ml (range 38.6–367.8) and 127.9 pg/ml (range 71.5–235.8), respectively. Median 24,25(OH)₂D₃ concentrations in cats with CKD and healthy cats were 9.8 ng/ml (range 2.5–32.5) and 8.8 ng/ml (range 1.2–38.4), respectively. There was no correlation between any vitamin D metabolite and age or serum creatinine concentration, and no correlation between 25(OH)D and hematocrit, blood pressure or urine protein:creatinine ratio.

Conclusions and relevance

Vitamin D metabolites were not significantly different in cats with CKD, which could have implications for recommendations regarding calcitriol supplementation.

Keywords: Calcitriol, cholecalciferol, calcifediol, chronic kidney disease-mineral and bone disorder

Introduction

Vitamin D is an essential nutrient used for various bodily functions. Vitamin D is most commonly known for its role in calcium-phosphorus homeostasis via the bone-parathyroid-kidney axis.¹ However, it also plays a significant role in immune regulation, inflammation, insulin secretion and hematopoiesis due to the various roles of the vitamin D receptor.^{2–4} In cats, vitamin D metabolism begins with dietary intake of vitamin D₃ (cholecalciferol) or vitamin D₂ (ergocalciferol). Cats lack the ability to synthesize vitamin D₃ via ultraviolet light, which is the primary mechanism in humans. After ingestion, cholecalciferol is metabolized in the liver to 25-hydroxyvitamin D (25[OH]D₃), which is then metabolized via 1- α (α)-hydroxylase to its active metabolite, calcitriol (1,25-dihydroxyvitamin D₃; 1,25[OH]₂D₃). Inactivation of 25(OH)D₃ by 24-hydroxylase

results in the production of 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃). Excretion of inactive vitamin D metabolites is accomplished by the biliary system.^{1-3,5}

Chronic kidney disease (CKD) is a common disease in cats characterized by progressively declining renal function, which affects upward of 50% of cats of all ages.⁶ CKD progression is associated with reduced glomerular filtration rate and the development of hyperphosphatemia, which in turn increases systemic concentrations of parathyroid hormone (PTH) and fibroblast growth factor-23 (FGF-23), all of which results in alteration in vitamin D metabolism.^{7,8} Proposed mechanisms of vitamin D metabolism alteration in CKD include decreased intake of cholecalciferol, decreased conversion of cholecalciferol in the liver to 25(OH)D₃, decreased 25(OH)D₃ activation via 1 α -hydroxylase, increased inactivation of 25(OH)D₃ and 1,25(OH)₂D₃ and loss of vitamin D binding protein in protein-losing nephropathy.^{1,3} These alterations in kidney function and mineral homeostasis as a result of CKD is generally referred to as CKD-mineral and bone disorder (CKD-MBD).

CKD-MBD has been extensively studied in humans and, to a lesser degree, in cats.⁵ To the authors' knowledge, there are no studies evaluating comprehensive vitamin D metabolite profiles of cats with CKD. The available literature primarily focuses on alterations in PTH, FGF-23 and 1,25(OH)₂D₃. Cats with uremic and end-stage CKD have lower concentrations of 1,25(OH)₂D₃ compared with healthy cats and cats with CKD.⁷ Other studies have shown no major difference in 1,25(OH)₂D₃ concentrations between healthy cats and cats with CKD.⁹ Recent studies have identified a novel vitamin D metabolite, a C-3, alpha-epimer of 25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), which is abundantly present in the plasma and serum of healthy cats.¹⁰ The exact role of this metabolite has not yet been established. Given its relatively high concentrations, it is important to elucidate whether this metabolite is altered in cats with CKD-MBD. The primary goal of this study was to evaluate the comprehensive serum vitamin D metabolite status in cats with various stages of CKD compared with healthy controls, including 25(OH)D₃, 3-epi-25(OH)D₃, 1,25(OH)₂D₃ and 24,25(OH)₂D₃. Our hypothesis was that cats with CKD would have lower 25(OH)D₃, 3-epi-25(OH)D₃ and 24,25(OH)₂D₃ concentrations compared with healthy cats and that 1,25(OH)₂D₃ concentrations would not differ between groups.

Materials and methods

Healthy cats

Serum samples were prospectively collected from cats in a healthy cat colony housed by Nestlé Purina Petcare Company (n = 13, IACUC #10389) as well as client-owned cats from The Ohio State

University College of Veterinary Medicine (OSU-CVM, n = 20, IACUC #2020A00000042-R1). All cats underwent a serum biochemistry profile, complete blood cell count (CBC), urinalysis and serum total thyroxine (TT4) testing. Healthy cats were further defined as having no relevant medical history, normal physical examination and creatinine below 1.6 mg/dl with urine specific gravity (USG) above 1.035.

Cats with CKD

Cats with International Renal Interest Society (IRIS) stages 1–4 were prospectively enrolled at OSU-CVM (n = 33). In addition, banked serum from cats with CKD that had been prospectively enrolled in a multicenter study at three veterinary teaching hospitals (OSU [n = 24, IACUC #2019A00000116], Tufts University Cummings School of Veterinary Medicine [n = 5, IACUC #G2020-61] and University of Pennsylvania Ryan Veterinary Hospital [n = 4, IACUC #806914]) was utilized. Cats were diagnosed with CKD based on persistent azotemia (creatinine >1.6 mg/dl) documented on at least two occasions and USG below 1.035, in combination with a history and physical examination consistent with CKD. All cats underwent Doppler blood pressure, serum biochemistry profile, CBC, urinalysis and TT4 testing at enrollment. Urine protein:creatinine ratio (UPC) was measured in 53 of the cats with CKD.

Cats with recent hospitalizations, evidence of acute kidney injury or changes in creatinine concentrations higher than 25% within 1 month before evaluation were excluded. Cats with uncontrolled systemic comorbid conditions were excluded. Use of vitamin D supplementation was not permitted but other concurrent therapies for CKD (eg, potassium supplementation, oral phosphate binders, anti-hypertensives, subcutaneous fluids) could be prescribed at clinician discretion.

Sample collection and vitamin D metabolite analysis

All cats were fasted before blood collection. After blood collection, samples were spun in serum separator tubes at $343\times g$ for 10 mins. Serum samples were stored at -80°C until batch submission to Heartland Assays, a vitamin D External Quality Assessment Scheme-certified laboratory, for vitamin D metabolite analysis via liquid chromatography-tandem mass spectrometry. Measured analytes included serum $25(\text{OH})\text{D}_3$, $3\text{-epi-}25(\text{OH})\text{D}_3$, $1,25(\text{OH})_2\text{D}_3$ and $24,25(\text{OH})_2\text{D}_3$.

Data analysis

Using preliminary data, an a priori sample size estimation was performed for an unpaired two-sample *t*-test with $\alpha = 0.05$ and 80% power with an allocation ratio of 2.0 (biomath.info/power/ttest.htm). Assuming a mean difference of nine units in serum 25(OH)D₃ concentration between groups (group means of 23 vs 32 ng/ml) and a common standard deviation of 14 ng/ml, a total of 89 subjects were required (30 in the healthy cat group and 59 in the CKD cat group).

Statistical analysis was performed using GraphPad Prism 10.4 (Graph Pad Software). Data were assessed for normality utilizing the D'Agostino and Pearson test and visual inspection of QQ plots. Vitamin D metabolites, clinicopathologic parameters and body condition score (BCS) were compared between groups using a Mann–Whitney U-test or Student's *t*-test (age, body weight) as appropriate. Correlations between vitamin D metabolites and clinicopathologic parameters, age and weight were assessed using Spearman's rank test.

Results

Of the 33 healthy cats enrolled in the study, domestic shorthair cats predominated ($n = 32$), with one domestic longhair cat. Sexes included spayed female ($n = 21$) and castrated male ($n = 12$). The median age of healthy cats was 8.8 years (range 1.7–17.3). Median body weight and BCS, using a 9-point BCS system, were 4.6 kg (range 2.8–6.5) and 6 (range 4–8), respectively. All healthy cats had a normal muscle condition score (MCS).

Of the 66 cats with CKD enrolled in the study, breeds included domestic shorthair ($n = 45$), domestic longhair ($n = 9$), domestic mediumhair ($n = 4$), two each of American Shorthair, Ragdoll and Siamese, and one each of Siberian and Tonkinese. Sexes included castrated male ($n = 36$) and spayed female ($n = 30$). The median age of cats with CKD was 13.9 years (range 1.4–20.4). Median body weight and BCS were 4.4 kg (range 2.8–7.6) and 5 (range 3–8), respectively. In total, 20 cats with CKD had a normal MCS. Muscle atrophy was noted to be mild, moderate or severe in 24, 15 and seven cats, respectively.

Pertinent renal parameters are reported in [Table 1](#). According to IRIS staging, cats had stage 1 ($n = 6$), stage 2 ($n = 36$), stage 3 ($n = 18$) and stage 4 ($n = 6$) CKD. Cats with CKD were significantly older, had significantly higher serum blood urea nitrogen and creatinine concentrations, and lower hematocrit (HCT) and USG than healthy cats ($P < 0.0001$ for all comparisons). There was no significant difference in body weight, BCS or serum phosphorus concentrations between groups. Vitamin D metabolite concentrations are shown in [Table 2](#). There were no significant differences noted in any vitamin D metabolite between the healthy cats and those with CKD ([Figure 1](#)), and

nor was there any correlation between any vitamin D metabolite with age or serum creatinine concentration. In addition, there was no correlation between 25(OH)D₃ and HCT, blood pressure or UPC in cats with CKD.

Table 1.

Concentrations of pertinent laboratory variables in cats with chronic kidney disease (CKD) and healthy cats

Variable	Cats with CKD	Healthy cats	<i>P</i> value
Hematocrit (%)	35 (21–46)	39 (33–55)	<0.0001
BUN (mg/dl)	39 (22–153)	21 (14–34)	<0.0001
Creatinine (mg/dl)	2.3 (1.2–7.7)	1.3 (0.8–1.5)	<0.0001
Phosphorus (mg/dl)	4.0 (2.7–11.5)	4.1 (2.7–5.3)	0.48
Total calcium (mg/dl)	9.9 (8.2–13.3)	9.9 (8.6–10.7)	0.43
USG	1.013 (1.007–1.050)	1.047 (1.038–1.069)	<0.0001

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Data are median (range). Laboratory reference intervals are not provided since multiple laboratories were utilized in this study

BUN = blood urea nitrogen; USG = urine specific gravity

Table 2.

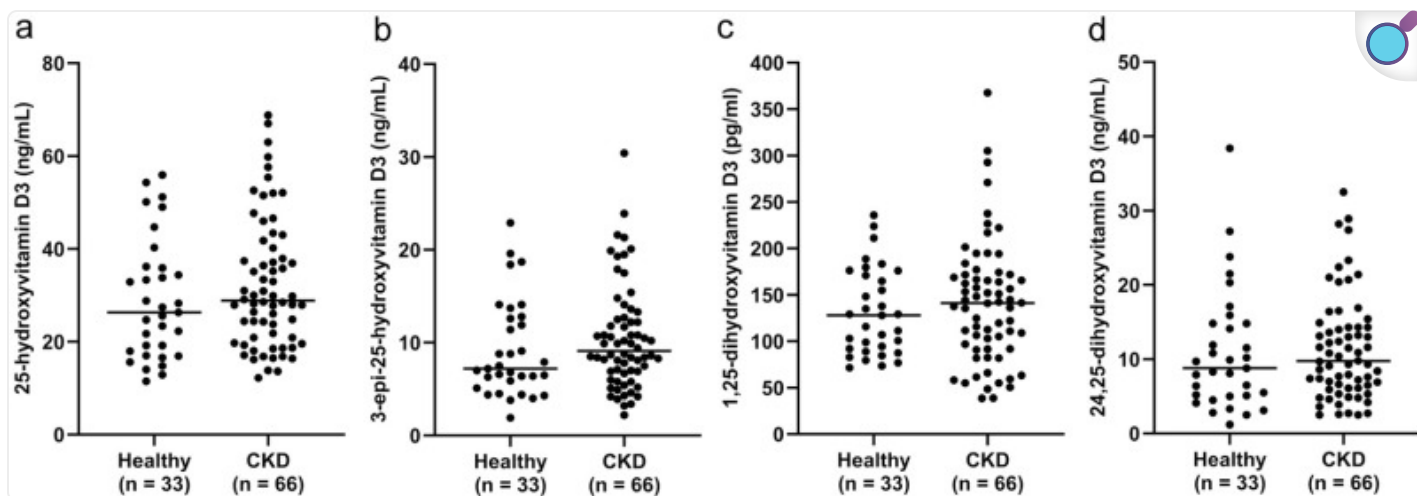
Concentrations of vitamin D metabolite concentrations, including 25-hydroxyvitamin D₃ (25[OH]D₃), 3-epi-25-hydroxyvitamin D₃ (3-epi-25[OH]D₃), 1,25-dihydroxyvitamin D₃ (1,25[OH]₂D₃) and 24,25-dihydroxyvitamin D₃ (24,25[OH]₂D₃) in cats with chronic kidney disease (CKD) and healthy cats

Vitamin D metabolite	Cats with CKD	Healthy cats	<i>P</i> value
25(OH)D ₃ (ng/ml)	28.9 (12.2–68.8)	26.3 (11.5–55.9)	0.23
3-epi-25(OH)D ₃ (ng/ml)	9.1 (2.2–30.4)	7.2 (1.9–22.9)	0.21
1,25(OH) ₂ D ₃ (pg/ml)	141.3 (38.6–367.8)	127.9 (71.5–235.8)	0.62
24,25(OH) ₂ D ₃ (ng/ml)	9.8 (2.5–32.5)	8.8 (1.2–38.4)	0.53

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Data are median (range)

Figure 1.



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Serum concentrations of (a) 25(OH)D₃, (b) 3-epi-25(OH)D₃, (c) 1,25(OH)₂D₃ and (d) 24,25(OH)₂D₃ in cats with chronic kidney disease (CKD) and healthy cats

It was not possible to determine absolute dietary vitamin D₃ (ie, cholecalciferol) intake in the cats with CKD given their variable appetites and variety of diets fed in rotation. Approximately 50% of cats with CKD were eating some or all of one or more veterinary therapeutic renal diets. The other cats were eating a variety of adult maintenance diets. Among the veterinary therapeutic renal diets, there was a wide range of cholecalciferol, approximately 17–94 IU per 100 kcal. Many cats with CKD were eating a variety of flavors of Purina Fancy Feast; the amount of cholecalciferol in the many flavors of Fancy Feast encompasses a wide range of 17–138 IU/100 kcal. All of the healthy cats were eating various adult maintenance diets that provided a range of approximately 16–25 IU cholecalciferol per 100 kcal.

Discussion

In this study, cats with CKD did not have significantly different serum vitamin D metabolites compared with healthy control cats. This differs from what has been demonstrated in humans and dogs with CKD, where most vitamin D metabolites are lower compared with healthy counterparts, becoming statistically significant by stage 3 in dogs with CKD.^{[11–13](#)}

Serum 25(OH)D₃ concentrations can be affected by a number of disease states and inflammation in cats;^{2,14,15} thus, it would not have been surprising for 25(OH)D₃ concentrations to be lower in cats with CKD if we consider CKD to induce a state of chronic inflammation. In addition, 25(OH)D₃ concentrations have been shown to be independently lower in humans with CKD and anemia.¹⁶ Despite the cats with CKD in this study having significantly lower HCT levels than healthy control cats, this did not seem to influence their 25(OH)D concentrations.

The serum 3-epi-25(OH)D₃ fraction made up approximately 25–30% of the total 25(OH)D₃ concentration in cats, consistent with what has been shown in cats with cardiomyopathy.¹⁷ In a study of humans with rheumatoid arthritis and CKD, serum 3-epi-25(OH)D₃ concentrations increased as severity of CKD progressed.¹⁸ In the cats included in our study, there was no correlation between this metabolite and serum creatinine concentrations. It is possible that the cats in this study were not affected severely enough to demonstrate a difference.

The 1,25(OH)₂D₃ (calcitriol) results in this study do mirror what has been previously demonstrated. Barber and Elliott⁷ demonstrated that calcitriol concentrations only decrease with end-stage CKD, and cats with earlier CKD had higher calcitriol concentrations compared with healthy cats. The majority of cats with CKD in this study had mild renal azotemia, with only six cats classified as having IRIS stage 4 CKD. It is possible that if more cats with late stage 3 and stage 4 CKD had been enrolled, a greater difference in vitamin D metabolite concentrations might have been demonstrated. Additional studies are necessary to assess vitamin D metabolites in cats with more advanced CKD.

Historically, calcitriol supplementation has been recommended for cats with CKD to address presumed renal secondary hyperparathyroidism. Potential adverse consequences can occur with calcitriol administration, including inducing hypercalcemia, hyperphosphatemia and an increased calcium-phosphate product, which increases the risk of soft tissue mineralization. Since calcitriol stimulates FGF-23 release, there may be additional nonspecific consequences of increasing calcitriol concentrations. Increased FGF-23 concentrations have been demonstrated in dogs receiving calcifediol and paricalcitol.^{19,20} We were unfortunately unable to analyze PTH and FGF-23 concentrations in this study.

There is a relationship between phosphate homeostasis and vitamin D metabolism. Increased serum phosphate should stimulate FGF-23 and decrease 1,25(OH)₂D₃ activation. Thus, it is possible that oral phosphate binders could have indirectly affected 1,25(OH)₂D₃ concentrations. However, only four cats with CKD received oral phosphate binders, and there was no significant

difference in serum phosphorus concentrations between groups, making this an unlikely contributor to the results.

The final metabolite we analyzed was $24,25(\text{OH})_2\text{D}_3$. It has been proposed that, in humans, serum $24,25(\text{OH})_2\text{D}_3$ concentration might serve as a reliable indicator of serum $25(\text{OH})\text{D}_3$ concentrations, and that the ratio of $25(\text{OH})\text{D}_3$ to $24,25(\text{OH})_2\text{D}_3$ might help predict relative 24-hydroxylase ($24[\text{OH}]ase$) activity, with decreased ratios indicating increased $24(\text{OH})ase$ activity and possible resistance to vitamin D treatment.²¹ There was no difference in this ratio between the cats with CKD and the healthy cats included.

Lastly, it seems unlikely that diet significantly influenced our results. There is not a large difference in the amount of cholecalciferol that feline adult maintenance diets provide compared with veterinary therapeutic renal diets. Most of the diets that cats were eating provided approximately 20–25 IU per 100 kcal, relative to the Association of American Feed Control Officials minimum requirement of 7 IU/100 kcal.²²

There were some limitations to this study. Some of the healthy cats included were from a research colony and may not reflect a client-owned cat population. We attempted to mitigate this by also enrolling client-owned healthy cats. Cats were not perfectly age-matched. In a study assessing $25(\text{OH})\text{D}_3$ status in cats with cardiomyopathy, every 1-year increase in age was associated with a 0.98 ng/ml decrease in $25(\text{OH})\text{D}_3$.¹⁷ We found no correlation between age and $25(\text{OH})\text{D}_3$ concentrations in this study. Similarly, a relationship between age and $25(\text{OH})\text{D}_3$ status has not been demonstrated in healthy adult cats²³ or healthy dogs or dogs with myxomatous mitral valve disease.^{24–26} In humans with CKD, there is no distinct correlation between age and $25(\text{OH})\text{D}_3$ concentrations.²⁷ It would have been ideal to enroll older healthy cats; however, it is challenging to identify cats aged over 13 years that do not have other medical conditions that can also influence vitamin D metabolism (eg, chronic enteropathy).¹

Dietary intake was not controlled, and nor could precise cholecalciferol intake be determined for all cats in this study. It is possible that hyporexia could have influenced decreased nutrient intake in cats with CKD. If this were to have affected results, we would have expected decreased vitamin D metabolites in cats with CKD, which we did not identify. In addition, in dogs, large doses of cholecalciferol supplementation have been required to affect serum $25(\text{OH})\text{D}_3$ concentrations,^{28,29} making it unlikely that standard amounts of ingested cholecalciferol significantly impacted these cats' $25(\text{OH})\text{D}_3$ concentrations.

Conclusions

Cats with CKD did not have significantly different vitamin D metabolite concentrations compared with healthy control cats. These findings illustrate that factors involved in CKD-MBD are unique to cats compared with humans and dogs. Calcitriol supplementation should be considered judiciously in cats with CKD, ideally based on comprehensive vitamin D metabolite and PTH concentrations.

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Footnotes

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Author note: The results of this study were presented in part as a research abstract at the American College of Veterinary Internal Medicine Forum 2024.

VJP and JMQ have received speaking honoraria and travel funds from Purina, Hill's, Royal Canin and Farmina. JMQ is an advisory board member for Nestlé Purina PetCare Company. LJR is currently employed by VCA Leesburg Veterinary Internal Medicine at The LifeCentre. These do not influence the results of the manuscript.

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Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals. Established internationally recognized high standards ('best practice') of veterinary clinical care for the individual patient were always followed and/or this work involved the use of cadavers. Ethical approval from a committee was therefore not specifically required for publication in *JFMS*. Although not required, where ethical approval was still obtained, it is stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective

studies). No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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